

Scientific Computing Project

Maxwell–Boltzmann Velocity Distribution

Problem Statement

The behavior of gases at the microscopic level is governed by the statistical motion of a large number of particles. In classical statistical mechanics, the velocities of particles in an ideal gas are not identical; instead, they follow a probabilistic distribution known as the Maxwell–Boltzmann distribution. This distribution plays a fundamental role in thermodynamics, kinetic theory of gases, and molecular physics, as it describes how particle speeds are distributed at thermal equilibrium.

The purpose of this project is to numerically simulate the velocity distribution of gas particles and compare the computational results with the theoretical Maxwell–Boltzmann distribution. Through this study, the student will investigate how random microscopic motion gives rise to predictable macroscopic statistical behavior.

For a gas in thermal equilibrium, the probability distribution of particle speeds is given by the Maxwell–Boltzmann distribution:

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}},$$

where:

- v is the particle speed,
- m is the mass of the particle,
- k_B is the Boltzmann constant,
- T is the absolute temperature.

The objective of this project is to computationally reproduce this distribution by generating random particle velocities, constructing the corresponding speed distribution, and analyzing how closely the numerical simulation agrees with theoretical predictions.

Project Overview

This project introduces the student to the foundations of computational statistical mechanics through numerical simulation and data analysis. Instead of solving deterministic equations of motion for individual particles, the project focuses on understanding the collective statistical behavior that emerges from large ensembles of randomly distributed particle velocities.

The student will simulate velocity components of gas particles using random sampling techniques and compute the resulting particle speeds. By constructing histograms of the simulated data and

comparing them with the theoretical Maxwell–Boltzmann curve, the project will demonstrate how probability distributions arise naturally in physical systems.

The project also provides an introduction to concepts such as Gaussian random variables, statistical distributions, histogram analysis, and numerical visualization. These ideas form the basis of many advanced topics in computational chemistry, molecular dynamics, and thermodynamic simulations.

Project Objectives

The objectives of this project are as follows:

- To simulate the velocities of gas particles numerically.
- To compute the corresponding distribution of particle speeds.
- To visualize the simulated distribution using histograms.
- To compare numerical results with the theoretical Maxwell–Boltzmann distribution.
- To study the influence of temperature on particle speed distributions.

Required Computational Tools

The implementation of this project should use the standard scientific Python ecosystem. Numerical computations and random number generation should be carried out using `NumPy`, while visualization and plotting should be performed using `Matplotlib`.

Students are encouraged to write modular and well-documented code, separating data generation, statistical analysis, and visualization into logically structured functions.

Project Tasks

The project consists of the following tasks.

1. Generation of Random Velocities

Generate random velocity components (v_x, v_y, v_z) for a large number of gas particles. The velocity components should follow Gaussian distributions consistent with thermal equilibrium.

The simulation should consider a sufficiently large number of particles to produce meaningful statistical results.

2. Computation of Particle Speeds

For each particle, compute the magnitude of the velocity vector using

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}.$$

Store the computed speeds for statistical analysis.

3. Construction of the Speed Distribution

Construct a histogram representing the distribution of particle speeds. The histogram should be properly normalized so that it approximates a probability density function.

The plot should include:

- labeled axes,
- appropriate bin selection,
- clear visualization of the distribution shape.

4. Comparison with Theoretical Distribution

Plot the theoretical Maxwell–Boltzmann distribution on the same graph as the simulated histogram.

Analyze the agreement between:

- the simulated numerical distribution,
- the theoretical probability distribution.

Discuss possible discrepancies arising from finite sampling and numerical approximation.

5. Temperature Dependence

Repeat the simulation for different temperatures and study how the velocity distribution changes with temperature.

Observe how increasing temperature affects:

- the average particle speed,
- the width of the distribution,
- the location of the distribution peak.

6. Statistical Analysis

Compute important statistical quantities such as:

- mean speed,
- most probable speed,
- root mean square speed.

Compare the numerical values with theoretical predictions from kinetic theory.

Suggested Code Structure

The project may be organized as follows:

```
maxwell_boltzmann_distribution/  
|  
|-- velocity_simulation.py  
|-- distribution_analysis.py  
|-- visualization.py  
|-- experiments.py  
|-- README.md
```

The program may include functions such as:

```
generate_velocities()  
compute_speeds()  
plot_histogram()  
plot_theoretical_distribution()  
run_temperature_experiments()
```

This modular structure helps maintain clarity and separation between simulation, analysis, and visualization components.

Expected Outcome

The simulation should produce a speed distribution that closely resembles the theoretical Maxwell–Boltzmann distribution for sufficiently large sample sizes. The histogram of simulated particle speeds should approach the smooth analytical curve predicted by statistical mechanics.

The project should also demonstrate how temperature influences the distribution of particle velocities. At higher temperatures, the distribution should broaden and shift toward larger speeds, reflecting the increase in average kinetic energy.

The comparison between numerical and theoretical results should provide insight into the statistical nature of thermodynamic systems and illustrate how macroscopic behavior emerges from microscopic randomness.

Extensions (Optional)

Students may further extend the project by exploring:

- two-dimensional velocity distributions,
- animation of moving particles,
- velocity distributions for different particle masses,
- comparison with non-equilibrium distributions,
- simulation of molecular collisions.

Deliverables

The completed project should include:

- well-structured and documented Python code,
- graphical visualizations of the simulated distributions,
- comparison plots between simulation and theory,
- a written explanation of the methodology and observations.

Learning Outcomes

Upon completion of this project, the student should have developed an understanding of:

- probability distributions in physical systems,
- Gaussian random sampling,
- histogram analysis and statistical visualization,
- numerical simulation of thermodynamic behavior,
- the connection between microscopic motion and macroscopic properties.

These concepts form an important foundation for advanced studies in statistical mechanics, computational chemistry, molecular simulations, and thermodynamics.